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NUISANCE ANIMAL TRAP

Abstract

A trap catches multiple animals and/or other nuisance animals in one setting. The trap may include a larger confinement volume having a one-way entry door on at least one side and in other embodiments on all sides that remain active no matter how many animals are in the trap. The entry door is designed such that once the animal has started through the door it cannot turn back. Access to remove the animals is also provided. A large bait station that can hold several days of bait and be easily rebaited with minimal contact with the trap is also provided. An inner trap may be contained within the outer trap to the bait for the target animals trapped in the outer trap.

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Background/Summary

BACKGROUND OF THE INVENTION [0001] This is claims priority to U.S. Provisional Patent Application Ser. No. 63/560,948, filed Mar. 4, 2024 and is related to U.S. Provisional Patent Application Ser. No. 63/332,717, filed Apr. 20, 2022, and U.S. Utility patent application Ser. No. 18/131,152, filed Apr. 5, 2023. Each of these prior applications is hereby incorporated by reference in its entirety.

[0002] This invention relates generally to animal traps, and more specifically, to an animal trap designed to capture iguanas, pythons and other nuisance species.

[0003] Pythons are a nuisance in many areas, including Florida because they threaten native wildlife and ecosystems. They are one of the most destructive invasive species in the Everglades. Pythons predate many native wildlife species and eat a wide range of wildlife, including mammals, birds, and other reptiles. They can even eat threatened or endangered species. Non-native Burmese pythons have established a breeding population in South Florida and are one of the most concerning invasive species in the area. Pythons compete with native wildlife for food, which includes mammals, birds, and other reptiles. Severe declines in mammal populations throughout Everglades National Park have been linked to Burmese pythons, with the most severe declines in native species having occurred in the remote southernmost regions of the Park where pythons have been established the longest. A 2012 study found that populations of raccoons had declined 99.3 percent, opossums 98.9 percent, and bobcats 87.5 percent since 1997 (Mammal Decline). Marsh rabbits, cottontail rabbits, and foxes effectively disappeared over that time (Marsh Rabbits Mortality). Pythons can reduce populations of native species, which can have cascading effects on other species and populations. Pythons may also prey on pets like cats and dogs. They may also carry invasive parasites that infect native snakes. The Florida Fish and Wildlife Conservation Commission's Python Action Team (PATRIC) and the South Florida Water Management District's Python Elimination Program work together to remove pythons from the wild. Additionally, the public is encouraged to remove and humanely euthanize pythons on private lands with landowner permission.

[0004] Hence, there is an ongoing need to have a system and method of controlling pythons and other nuisance species.

SUMMARY OF THE INVENTION

[0005] A system and method for trapping and containing pythons, iquanas and other nuisance species are provided by various embodiments of this invention.

[0006] This invention is a trap according to various embodiments and is designed to catch multiple animal's in one setting. The features of the trap in various embodiments include a large box or cage consisting of one-way door(s) or openings on at least one side and in other embodiments all sides that never lock and remain active no matter how many animals are in the trap. An entry door is provided and designed such that once the animal has started through the door it cannot turn back. Access to remove the animals is also provided. A large bait station that can hold several days of bait/food and be easily rebaited while minimal contact with the trap is also provided.

[0007] In various embodiments, this invention is directed to a trap for a target animal such as a python and utilizes first and second traps, one contained within the other. The inner trap may include bait for mice and rodents which are to be used as bait animals. Once the rats or mice as bait animals are in the inner trap, the pythons are attracted to the inner trap and enter the outer trap via a one-way door. Once in the outer trap, the pythons cannot exit the outer trap nor access the bait animals in the inner trap.

[0008] The trap in various embodiments may be designed to be easily disassembled for easy transportation and shipping. The trap may have one or more anchor points so it can be securely

anchored to prevent the trap from being stolen or blown away in bad weather such as a hurricane. The access door may have a lock to prevent unauthorized access to the trap and its contents. The trap is designed to operate quietly with no slamming doors or other components that would cause noises that would scare the animals away.

[0009] Pythons are an invasive species to the US and are destroying environment and wildlife in south and central Florida. As it turns out, animals such as iguanas are part of the python's diet. A trap full of animals such as iguanas will attract pythons. The design of the trap according to various embodiments of this invention will allow a python to enter in the same fashion as the animal without restriction. Once a python enters the trap, it will be in the same predicament as other animals with no way out. Pythons have now started to move out of the everglades in search of food, such as dogs, cats, rats, racoons, and animals. This invention could have a cover made from canvas or like kind material to be used as camouflage and to conceal/hide the trap from view. In winter months it would hold heat in the trap further attracting the animals in search of warm places when it gets colder outside. The cover can be in a variety of colors or color combinations to best suit the situation/environment.

[0010] Other embodiments of this invention include an escape door for non-targeted species, such as fur bearing species, to escape the trap should they become captured. Still further embodiments of this invention include a bait compartment and heating pad to further entice the targeted animals to enter the trap.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] The above-mentioned and other features and advantages of this invention, and the manner of attaining them, will become more apparent and the invention itself will be better understood by reference to the following description of embodiments of the invention taken in conjunction with the accompanying drawings. Like reference numerals are used to indicate like parts throughout the various drawing figures, wherein:

[0012] FIG. 1 is a perspective view of one embodiment of the trap according to this invention;

[0013] FIG. 2 is a perspective view of the base of the trap of FIG. 1;

[0014] FIG. 3 is a perspective view of the side and top walls of the trap of FIG. 1 in a generally planar configuration;

[0015] FIG. 3A is a view similar to FIG. 3 with only an access door on the top wall shown;

[0016] FIG. 4 is a perspective view of the top and side walls being assembled into an erected configuration;

[0017] FIG. 5 is an enlarged perspective view of a corner of the base of the trap being secured to an underlying ground surface;

[0018] FIG. 6 is an enlarged perspective view of a portion of the erected side walls being coupled to the base;

[0019] FIG. 7 is a side elevational view of the assembled trap according to one embodiment of this invention with bait receptacles shown in phantom lines;

[0020] FIG. 8A is an enlarged side elevational view partially in cross-section showing an animal approaching tines of an entry door on the base of the trap;

[0021] FIG. 8B is a view similar to FIG. 8A with the animal entering the entry door and deflecting the tines inwardly;

[0022] FIG. 8C is a view similar to FIGS. 8A and 8B with the animal trapped in the trap;

[0023] FIGS. 9A-9F are alternative embodiments of the tines of the entry door according to various embodiments of this invention;

[0024] FIG. 10 is a perspective view of an auxiliary cage according to various embodiments of this

invention;

[0025] FIG. **10A** is a view similar to FIG. **10** showing handles and a trap door on the auxiliary cage;

[0026] FIG. **11** is a perspective view of the erected side and top walls being assembled with a base of the auxiliary cage;

[0027] FIGS. **12A** and **12B** are sequential side elevational views partially in cross-section of the auxiliary cage being coupled to the trap according to one embodiment of this invention;

[0028] FIG. **13** is a perspective view of the coupled trap and auxiliary cage of FIGS. **12A-12B**;

[0029] FIG. **14** is a perspective view of an alternative embodiment of the trap according to this invention;

[0030] FIG. **15** is an enlarged view of a portion of the trap of FIG. **14**;

[0031] FIG. **16** is a perspective view of a further embodiment of this invention including an escape door for non-targeted captured species;

[0032] FIG. **17** is a top plan view of the trap of FIG. **16**;

[0033] FIGS. **18A-18B** are sequential enlarged side elevational views of the escape door of FIGS. **16-17**;

[0034] FIG. **18C** is a view similar to FIG. **18A** of another embodiment of the escape door;

[0035] FIG. **18D** is a still further embodiment of the escape door;

[0036] FIG. **19A** is a front elevational view of one embodiment of the escape door;

[0037] FIG. **19B** is a view similar to FIG. **19A** of a still further embodiment of the escape door;

[0038] FIG. **19C** is a view similar to FIGS. **19A-19B** of a yet further embodiment of the escape door;

[0039] FIG. **19D** is a view similar to FIGS. **19A-19C** of a further again embodiment of the escape door;

[0040] FIG. **20** is a perspective view of the trap being positioned on a pad according to one embodiment of this invention;

[0041] FIG. **21** is a side elevational view of a rat trap which may be used as a stand-alone trap or in combination with a trap intended for animals or other animals as disclosed herein;

[0042] FIG. **22** is an exploded view of a removable trapping/bait box from the embodiment of FIG. **21**;

[0043] FIG. **23** is a top elevational view of the embodiment of FIG. **21**;

[0044] FIG. **24** is a side elevational view of a bait cone which may be used in conjunction with the embodiment of FIG. **21**;

[0045] FIGS. **25A** thru **25D** are various views of alternative embodiments of the rat trap;

[0046] FIG. **26** is a perspective view of a trap within a trap embodiment of this invention;

[0047] FIG. **27** is a view similar to FIG. **26** with an outer trap shown in phantom lines to allow for viewing of an inner trap; and

[0048] FIG. **28** is a side elevational view of the embodiment of FIGS. **26-27** showing the bait and target animals therein according to one embodiment of this invention.

DETAILED DESCRIPTION OF THE INVENTION

[0049] With reference to the drawings, one embodiment of a trap **10** according to this invention is shown in FIG. **1**. The trap **10** is described herein for use to trap iguanas or pythons, but one of ordinary skill in the art will appreciate that this invention is not limited to use for such animals only, but is readily applicable for trapping any animal or animals **12**. According to the embodiment shown in FIG. **1**, the trap **10** has four side walls **14** and a top wall **16**, but the trap **10** may be in other configurations according to this invention. Each wall **14**, **16** may be constructed of a wire mesh **18** formed of a number of intersecting wires **20**. The walls **14**, **16** may be of another construction according to this invention.

[0050] As shown in FIG. **3**, each side wall **14**, may be pivotally joined to the top wall **16** by a number of clips **22** which join a perimeter edge of each side wall **14** to one of the perimeter edges

of the top wall **16** to form a cruciform arrangement. Each side wall **14** may be pivoted downwardly about the associated clips **22** to be generally perpendicular to the top wall **16** and the adjacent side walls **14** as shown in FIG. 4. The adjacent edges of the adjacent side walls **14** may be joined together by twisting, turning, screwing or threading a multi-turn, convolute elongate wire **23** around the perimeter wires of each of the adjacent side walls **14**. The top wall **16** is shown with an access door **24** pivotally mounted to the top wall **16** and secured in a closed position by a clasp **26**. In alternative embodiments, the trap **10** may be manufactured in one piece with all of the side walls permanently affixed to each other as well as a base **28**.

[0051] The trap **10** according to various embodiments includes a base **28** as shown in FIG. 2. The base **28** is assembled with the side and top walls **14**, **16** to form the trap **10**. The base **28** includes a bottom panel or wall **30** which may be mesh or of another configuration. A number of walls **32** extend upwardly from the bottom panel **30** equal in number to the number of side walls **14** on the trap **10**. In the embodiment shown in FIGS. 1-2, each wall **32** of the base **28** includes an entry door **34** for the animal **12** to enter the trap **10**. Each entry door **34** may extend substantially the full length of the associated wall **32** as shown in FIG. 2.

[0052] As shown most clearly in FIGS. 2 and 5-6, the base **28** includes corner posts **36** at each juncture between adjacent walls **32**. A base frame **38** is formed around the bottom perimeter of the base **28** and each frame member **40** of the base frame **38** may be L-shaped in cross-section. A gusset **42** at each corner of the base frame **38** may have an aperture **44** through which a stake **46** or other anchor post may be inserted to at least temporarily secure the trap **10** to the ground surface there beneath. An upper rim **48** of each wall **32** may have a spaced double wall **48a**, **48b** construction as shown in FIGS. 5-6. Each rim wall **48a**, **48b** of the upper rim **48** may have a number of holes **50** therein aligned with a complimentary hole **50** in the adjacent rim wall **48a**, **48b**.

[0053] As seen in FIG. 6, a lower edge of each side wall **14** may be seated on the upper rim **48** between the rim walls **48a**, **48b** to mate the base **28** with the side walls **14** and top wall **16** of the trap **10**. Mechanical fasteners, screws or bolts **52** may be inserted into the aligned holes **50** to secure the side walls **14** to the base **28**.

[0054] Each entry door **34** may have a number of spaced and downwardly oriented tines **54** suspended from a hub **56** mounted on a rod **58**. Each rod **58** is mounted between a pair of end brackets **60** and spaced from the upper rim **48** of the associated base wall **32**. Each tine **54** is mounted via its associated hub **56** for pivotal movement about the rod **58**. As shown in FIG. 5, each tine **54** has a spike **60** or other end feature at a distal end of the tine **54**. The spike or end feature **60** of each tine **54** is juxtaposed to an inner surface of an upstanding leg or detent **62** of the frame member **40** of the base frame **38** such that the tine **54** may only pivot inwardly into an interior of the trap **10**. Each tine **54** of each entry door **34** is prevented by the upstanding leg **62** of the frame member **40** from pivoting outwardly beyond the upstanding leg **62**. Each tine **54** is biased by gravity to rest against the associated upstanding leg **62** since the rod **58** is positioned outwardly from the upstanding leg **62** as shown in FIGS. 6, 7, 8A and 8C. Alternative embodiments of the trap **10** may include one or more torsion springs biasing the tines **54** outwardly against the upstanding leg **62**.

[0055] As seen in FIG. 7, one or more bait receptacles **64** may be suspended from the top wall **16** and/or located on the bottom wall **30** to entice the animal **12** to enter the trap **10** via one of the entry doors **34**. Referring to FIGS. 8A thru 8C, the animal **12** may approach one of the entry doors **34** in search of the bait **64** within the trap **10** (FIG. 8A). The animal **12** pushes through one or more tines **54** of the entry door **34** thereby pivoting the associated tines **54** inwardly about the rod **58** from which the tines **54** are suspended. The tines **54** offer little resistance to the animal **12** entering the trap **10** in search of the bait **64** (FIG. 8B). Once the animal **12** has entered the trap **10**, each inwardly pivoted tine **54** which allowed entry of the animal **12** returns to its rest position by gravity or otherwise against the upstanding leg **62** of the frame member **40** as shown in FIG. 8C. The

animal **12** is prevented from escaping the trap **10** by pushing the tines **54** of the entry doors **34** outwardly because the tines **54** are prevented from pivoting outwardly beyond the upstanding leg **62** of the frame member **40**. The spacing between the tines **54** is sufficiently small to prevent a trapped animal **12** or other animal from escaping the trap **10** between the tines **54**. The hub **56** of each tine **54** is juxtaposed against the adjacent hub **56** of the adjacent tine **54** to prevent lateral movement of the tines **54**. Moreover, other animals **12** may also enter the trap **10** via the entry doors **34** in a similar manner even though one or more animals are already trapped in the trap **10**. Each tine swings independently so as the animal enters the trap the tines slide/hug along its body, so as they enter the trap they don't create an opening large enough for an animal that is already in the trap to escape.

[0056] In alternative embodiments of this invention, a live bait compartment collects live bait that will attract the target animals or reptiles. A full rodent may be within the trap **10**. The rodent trap attracts rats or other bait that will be live captured in the bait compartment. Rodents are a favorite food of these reptiles. Rats are also a nuisance pest and plentiful making it an ideal bait. A self-baiting trap may be located deep in any portion of the trap **10** to draw the target bait deep into the trap. No access is provided for the reptiles or non-targets to get to the rodents in the bait compartment to eat them to provide an endless source of live bait for the target animals with the trap **10**. Another option is to provide a compartment that could be preloaded with rats for bait according to this invention.

[0057] Alternatively, a separate rodent trap that can be easily added to the trap **10** may be a freestanding trap for inner-city use.

[0058] In various embodiments of this invention, the tines **54** are constructed to allow non-target animals to escape (raccoons, opossums, cats, etc.). Non-target animals may have front paws with much greater dexterity for holding food or other smaller tasks. The animals have large spread out toes more conducive to climbing than for finer dexterity. This unique difference allows non-target animals to grab the tines in ways the animal cannot. Non-target animals may simply grab the tine and pull it up to create an exit. Pythons and other snakes, for example, have no hands, but like animals they push their way into the trap and remain captured as they have no way to open the tines.

[0059] Alternative embodiments of the entry door **34** and/or end feature **60** are shown in FIGS. **9A** thru **9F** according to this invention. Each tine **54** may have an additional spike **60a** projecting into the interior of the trap **10** as shown in FIG. **9A** to deter a trapped animal **12** from exiting the trap **10**. A pivot plate **60b** as in FIG. **9B** may be positioned relative to one or more tines **54** to block the associated tine(s) **54** from pivoting inwardly when stepped on by the animal **12** trapped in the trap **10**. An end of the pivot plate **60b** pivots upwardly in the direction of arrow **A** to block inwardly movement of the associated tine(s) **54**. The end feature **60c** of the tine **54** may be an inwardly directed flange **60c** such that when the trapped animal **12** steps on the flange **60c**, the tine **54** is forced against the upstanding leg **62** to prevent the animal **12** from exiting through the entry door **34**. The end feature **60d** may be one or more spikes **60d** projecting from the flange **60c** and/or a shaft of the tine **54**.

[0060] A barricade **61** may be added to the L-shaped frame member and spaced from the upstanding leg **62** as shown in FIGS. **9E-9F**. The barricade **61** may have one or more peaks **60e** formed therein with a groove **66** between each peak **60e**. The groove **66** may be laterally oriented as shown in FIG. **9F** to allow for lateral movement of the tine **54** seated in the groove **66**.

[0061] Another aspect of various embodiments of this invention is shown in FIGS. **10-10A** in the form of an auxiliary cage **70** which may have four side walls **72** and a top wall **74**. Upper edges of the side walls **72** may be joined to perimeter edges of the top wall **74** with clips **22** as in the trap **10**. One or more of the side walls **72** may include a mirror **72a** to provide a reflection of any animal **12** within the cage **70** or trap **10**. The side wall **72b** opposite from the mirror **72a** may be shorter than the other side walls **72**. Adjacent side edges of the side walls **72** may be joined to each other with

convolute wires **23** as in the trap **10**. The top wall **74** may have a trap door **76** which is mounted on a pivot shaft **78** to divide a trap door wall **80** into major and minor portions **80a**, **80b**. The minor portion **80b** may have a weight **82** to bias the trap door wall **80** into a closed position about the pivot shaft **78**. When an animal **12** or other animal steps on the major portion **80a** of the trap door wall **80**, the mass of the animal **12** or other animal overcomes the mass of the weight **82** and the trap door wall **80** pivots about the pivot shaft **78** with the major portion **80a** pivoting downwardly and the animal **12** falling into the auxiliary cage **70**. Once the mass of the animal **12** is removed from the major portion **80a** of the trap door wall **80**, the weight **82** pivots the minor portion **80b** downwardly to the closed position shown in FIGS. **10-11**. The trap door **76** may be included in the auxiliary cage **70**, the trap **10** or both of them in various embodiments of this invention.

[0062] The auxiliary cage **70** is easily transported by a user grasping one or both cable handles **84** attached to the side walls **72** and/or top walls **74**. The cable handles **84** are located on the exterior of the auxiliary cage **70** so that any animal or other animal **12** within the cage **70** may not damage, weaken or interfere with the cable handles **84**. The auxiliary cage **70** includes a base **86** having a wire mesh **18** and a perimeter frame **88**. Three sides of the perimeter frame **88** include spaced double wall **88a**, **88b** construction as shown in FIG. **11**. Each rim wall **88a**, **88b** of the frame **88** may have a number of holes **50** therein aligned with a complimentary hole **50** in the adjacent frame wall **88a**, **88b**. As seen in FIG. **11**, a lower edge of each side wall **72** may be seated on the frame **88** between the walls **88a**, **88b** to mate the base **86** with the side walls **72** and top wall **74** of the auxiliary cage **70**. Mechanical fasteners, screws or bolts **52** may be inserted into the aligned holes **50** to secure the side walls **72** to the base **86**.

[0063] One side of the base frame **88** may have an entry portal **90**. The entry portal **90** is formed by an L-shaped member **92** on the fourth side of the frame **88** having an upstanding leg **92a**. Side posts **94** extend upwardly from the frame **88** and a U-shaped upper member **96** extends between the posts **94** and spaced above the L-shaped member **92**. Each side post **94** also has a U-shaped cross-section. The spaced walls of the U-shaped members **94**, **96** may have a number of holes **50** therein aligned with a complimentary hole **50** in the adjacent member. As seen in FIG. **11**, a lower edge of the shorter side wall **72b** may be seated on the upper member **96** between the spaced walls to mate the base **86** with the side walls **72** and top wall **74** of the auxiliary cage **70**. The lower ends of the side edges of the walls **72** adjacent to the shorter wall **72b** may be seated between the walls of the U-shaped side posts **94**. Mechanical fasteners, screws or bolts **52** may be inserted into the aligned holes **50** to secure the side walls **72** to the base **86**. The posts **94**, L-shaped frame member **92** and the upper member **96** define the entry portal **90** of the auxiliary cage **70**.

[0064] When the trap **10** has one or more animals **12** captured therein, the auxiliary cage **70** may be coupled to the trap **10** to off-load the animals **12** into the cage **70**. One or more brackets **98** with an upward U-shaped slot **100** may extend from a common side of the trap **10** as shown in FIG. **12A**. The tines **54** of the entry door **34** on the side of the trap **10** with the brackets **98** may be positioned outwardly of the upstanding leg **62** and hang generally vertical from the rod **58**. A brace **102** is provided on the posts **94** extending between the walls of the post **94**. The cage **70** is coupled to the trap **10** by lowering the cage **70** and positioning the brace **102** to be seated in the U-slot **100** of the bracket **98** as shown by arrow B in FIG. **12A**. Alternative embodiments of this invention may have the bracket on the auxiliary cage and the brace on the trap or another attachment system. The tines **54** of the entry door **34** adjacent to the cage **70** and the entry portal **90** are positioned against the upstanding leg **92a** of the L-shaped frame member **92** as shown in FIG. **12B**. In this arrangement, the cage **70** is coupled to the trap **10** and the entry door **34** is in communication with the entry portal **90** to allow animals **12** in the trap **10** to pass through the entry portal **90** and into the cage **70**. The mirror **72a** is visible by the animals **12** in the trap **10** and the reflections seen in the mirror **72a** entice the animals **12** to pass through the entry door **34** and entry portal **90** to enter the cage **70**. The animals **12** in the cage **70** are prevented from exiting the cage **70** via the entry portal **90** by the tines **54** if the trap **10** juxtaposed to the upstanding leg **92a** of the cage frame member **92**.

[0065] The auxiliary cage **70** mated with the trap **10** is shown in FIG. **13**. The animals **12** in the trap **10** may be off-loaded by migration into the cage **70**, which may then be decoupled from the trap **10** for relocation of the animals **12** in the cage **70**. The off-loading of the animals **12** in the trap **10** is accomplished without moving the trap **10** according to various embodiments of this invention. The entry portal **90** of the cage **70** may be temporarily blocked by bungee cords stretched across the entry portal **90** and hooked to the mesh **18** of the cage **70** or another means while transporting the animals **12** in the cage **70**. A gusset **42** at each corner of the base frame **38** may have an aperture **44** through which a stake **46** or other anchor post may be inserted to at least temporarily secure the trap **10** to the ground surface there beneath.

[0066] The trap **10** may have any number of entrances and holding chamber entrances. An additional aspect of various embodiments of this invention is that the trap **10** does capture the target species (i.e. animals) but does not retain “non-target species” (i.e., raccoons, possums, cats, etc.). This is likely because the non-target species have paws (hands) that are different from the animal's, and they are able to grab the tines (doors) and open them to escape.

[0067] In alternative embodiments of this invention, the auxiliary cage **70** may be connected to the trap **10** with a chute or tunnel that extends from the trap **10** into the auxiliary cage **70**. The tunnel or chute may house the tines **54** and provide the alignment for connecting the trap **10** and the auxiliary cage **70**. Traps **10** without the auxiliary cage **70** connected thereto will instead have another entrance.

[0068] Another aspect of some embodiments is the addition of one or more water bowls and filler bottles the trap **10**. This allows for the possibility that the animals could be in the trap for up to a week and although they can survive without any issues, providing water is often considered more humane. The bowls can also collect rain and sprinkler water. When an animal could be in the trap for extended periods (a few days), it's only humane to provide water. Various embodiments according to this invention may include a water bowl that is affixed to the trap so it can't be knocked over, a removable and/or refillable water bottle that attaches to the side of the trap with a drip feeder, or both.

[0069] An alternative embodiment of the trap **10** according to this invention is shown in FIGS. **14-15** in which the confinement volume for the animal(s) **12** is defined by the trap **10** and is in communication with a chute **104** with the entry door **34** found in a free end of the chute **104**. The chute **104** is in communication with the confinement volume of the trap **10**. While the chute **104** and associated entry door **34** is located on one side of the trap **10**, it will be understood by those of ordinary skill in the art that entry doors **34** and/or chutes **104** may be found on as many sides of the trap **10** as desired.

[0070] The tines **54** of the entry door **34** of the embodiment in FIG. **15** are formed in double width plates **54a** with apertures **106** in each plate **54a**. Each tine **54** is independently pivotal from the rest position shown in FIG. **15** relative to the other tines **54** in the entry door **34**. Each tine **54** is pivotally mounted to an upper rung **108** at the free end of the chute **104** in this embodiment. Multiple entry doors **34** may be spaced apart from one another as shown in FIG. **15**. The tines **54** of the trap **10** of FIGS. **14-15** when in the closed configuration as shown rest against a detent which prevents them from pivoting outwardly and allowing the captured animals from escaping the trap **10**. The detent which prevents outward pivotal movement of the tines **54** may be the bottom of the chute **104** (FIG. **15**) or another distinct obstruction or component of the trap **10**. As is shown in FIGS. **14-15**, the tines **54** when in the rest or closed position are not vertical but canted or inclined relative to the horizontal bottom of chute **104**.

[0071] In various embodiments, the tines **54** are at about a 25 degree angle minimum facing in, for uneven ground. If the trap with doors on an uphill inclination, the tines **54** would hang open enough to allow for the animals to escape. With the tines **54** at the appropriate angle, the tines **54** create a “wedge” against the base of the trap **10** and don't require any component on the base to stop them from swing outwardly.

[0072] The tines **54** may act and pivot independently from each other, to conform to the shape or profile of the animal as they enter the trap **10** making to more difficult for the animal to reverse course or escape from the trap **10**. The individual tines **54** according to various embodiments are removable for repair and/or replacement as needed.

[0073] In alternative embodiments of this invention, the side walls **14** may contain all the tines **54** which are attached directly to the mesh of the side walls **14** and are removable for repairs. The side walls **14** may connect directly to the base **28** which may be flat. The auxiliary cage **70** may have a similar construction. The auxiliary cage **70** may have a receiver for the chute/tunnel from the main trap **10**, the water features, a locking door where the receiving chute enters so that when the auxiliary cage **70** is removed from the trap **10**, the doors closes and locks to stop the animals from escaping and then a locking access door on top.

[0074] Still further embodiments of the trap **10** according to this invention are shown in FIGS. **16-20**. The embodiments of FIGS. **19-20** include features which allow for non-targeted animals or species which enter the trap **10** to escape while containing the target animals and species in the trap **10**. The trap **10** of various embodiments of this invention is specific to targeting reptiles that are omnivores and carnivores. However, non-targeted animals or species may become captured in the trap and various embodiments of this invention allow for such non-targeted animals such as fur bearing, non-target animals to escape the trap **10**. Such features further increase the likelihood that any fur bearing, non-target animal will escape, which is important for wide spread adoption and usage of the trap **10**. Such features may be incorporated into the trap **10** because of bait changes necessary to attract target animals that will also increase interest from non-target or fur bearing animals and the likelihood that they will go deeper in to the trap. It has been observed that fur bearing animals (i.e., raccoons, foxes, opossums, etc.) once they realize they are trapped will begin pulling on the sides of the trap. On the other hand, reptiles, especially snakes, will start pushing on the sides of the trap.

[0075] FIGS. **16-19D** show various embodiments of an escape door **110** included in a portion of the trap **10**. The escape door **110** may be incorporated into any wall **14**, **16** of the main trap **10** or any wall or top of a holding chamber **112** in communication with the main trap **10** and/or the auxiliary cage **70**. The escape door **110** may be added to the holding chamber **112** of the trap **10** according to various embodiments and configurations.

[0076] The escape door **110** may be a simple, flat member that will only open inwardly, one example is shown in FIG. **16** with the escape door on a side wall **114** of the holding chamber **112**. The escape door **110** may include a stop **111** projecting upwardly from a distal end of the door **110** opposite a pivotal connection **116**. The door **110** may be located on any portion of the trap **10** or the holding chamber **112**, including the side wall and/or top wall. One end of the door **110** is coupled to the side wall **114** by a pivotal connection **116**. The door **110** may be biased into a closed configuration generally flush with the remainder of the side wall **114** when the stop **111** contacts the side wall **114** as shown in FIG. **16** by a biasing member **118** such as a spring, elastic member or bungee cord. One end **118a** of the biasing member **118** may be secured to the escape door **110** and the opposite end **118b** may be secured to an adjacent portion of the side wall **114**. As the non-targeted or fur bearing animal captured in the holding chamber **112** is searching for an exit to escape, they will eventually come across this escape door **110** and when they pull on the escape door **110** from inside the chamber **112**, the door **110** will pull open, arrow B, by stretching the biasing member **118**, providing an exit, and the escape door **110** will snap close by the tension provided by the biasing member **118** as the non-target animal exits the chamber **112**. In the embodiment of FIG. **18A**, the end **118a** of the biasing member **118** is connected to the middle region of the escape door **110**. In the embodiment of FIG. **18C**, the escape door **110** includes an extension member **120** extending perpendicularly from the middle region of the outer face of the escape door **110** with the biasing member **118** traversing over a distal end of the extension member **120**. This arrangement creates more leverage and tension on the door **110** to make it harder to open

requiring more force to open.

[0077] In the embodiment shown in FIG. **18D**, a handle **122** is added to the escape door **110**. The handle **122** extends perpendicularly from an inner face of the door **110** at the free end of the door opposite from the pivotal connection **116** of the door **110** to the remainder of the trap **10**. The handle **122** provides easier manipulation and opening of the door **110** for a non-targeted animal to open the door **110** and escape the trap **10** or holding chamber **112** as the case may be. The handle **122** provides the fur bearing trapped animal with something additional to grab on to and more leverage to get the door **110** open.

[0078] The biasing member **118** according to various embodiments of this invention may be located and/or attached to different locations on the door **110** itself and/or the trap **10** or holding chamber **112**. For example, with end **118a** attached to the side wall **114** proximate the pivotal connection **116**, the end **118b** of the biasing member may be attached to the center of the door **110** (see FIG. **19A**), it may be attached to the sides of the door **110** (see FIG. **19B**) or it may be attached to the top of the door **110** (see FIG. **19C**) and attach to the top of the door. Alternatively, the end **118b** of the pivotal connection **118** may be attached to the trap **10** or holding chamber proximate the free end of the door **110** and the end **118a** attached to the free end of the door **110** (see FIG. **19D**).

[0079] A further feature of embodiments of this invention is shown in FIG. **20** is a thermal base **124** underneath the trap **10**. The thermal base **124** may be integral to the trap **10** or an underlayment on which the trap **10** is supported. The base **124** may be a polymer or styrene composition that is black and absorbs heat from the daytime sun and provides a heat source at night. A black or dark color absorbs the heat and a thicker the base **124** will have greater thermal retention capability. Many of the target reptiles/snakes are nocturnal and are attracted to heat on cool nights. The rodents for bait are also nocturnal. The additional heat source will only help attract both, bait and target animals.

[0080] In alternative embodiments, the base **124** may have a direct heat source, such as one or more candle, a heat source such as that used in pocket/hand warmers or a camping/tent heater. A solar panel, battery, photocell/timer, heat source could also be included. The heat could come from a heat lamp or a radiant source, like heated rocks. Such additional heating sources could be located in any chamber of the trap **10**. This would allow the trap **10** to be located just about anywhere and provide a heat source for these target reptiles. It is not uncommon to for these target reptiles/snakes to lay at night on roads heated by the sun during the daytime in an attempt to stay warm. During cool nights and winter months this feature could be more attractive to the target animals than actual bait.

[0081] Further alternative embodiments of this invention are shown in FIGS. **21** through **25D** which are generally directed to a bait trap feature which may be a stand-alone trap or used in conjunction with the trap **10** for a self-baiting trap feature. In certain embodiments, the bait trap assembly **130** is directed to a rat only trap using the top portion (mechanicals) of the trap **10**. Historically, (for the last 30 years) commercially speaking, rats have been poisoned using "bait stations" where the rat passes thought the bait station smells the poison and consumes a dose of poison that will kill it in the rear future. Issues with such bait stations include (1) over the years, the rats that didn't consume a lethal dose began to build immunity to the poison; (2) the dose of poison cannot be increased as it will have a negative effect on the food chain, as the dead rats can be consumed by other animals (cats, foxes, raccoons, etc.) and thus risk killing them also. It has been observed that the rat populations are exploding.

[0082] One solution to this situation is to live trap the rats and exterminate them in a controlled setting. The conventional spring loaded rat kill traps are not realistic for large scale use. Rats need to be caught in bulk for effective management. The rat trap assembly **130** (see FIGS. **21-23**) builds off the design of the rat trap add-on for the animal trap **10** and may be used as a stand-alone trap or in combination with the animal trap **10**. Simply put, the rat trap assembly includes a base **132**. The base **132** is much larger and sloped to provide a gradual ramp to the entrance(s) **134** of the trap assembly **130**. Most bait stations are designed to sit close to the buildings and walls because

rodents like to have their whiskers touching the something such as the building walls as they are move about in solitude because the tactile whisker sensation makes them feel safer. One aspect of the rat trap assembly **130** in various embodiments is the ramped base **132** at each end so it will work from either direction while sitting against a wall so the rat continues to feel safe.

[0083] As the rat approaches the trap assembly **130** it will smell the bait to further attract them. The top portion **136** of the assembly **130** is fully removable and is securely fastened to a trapping chamber **138** when in use so other predators can't get in or open the trap assembly **130**. The trapping chamber **138** may be staked, weighted, or otherwise secured as well. The trapping chamber **138** may hold hundreds of rats. When it is time to process or empty the trap assembly **130**, the top portion or mechanicals **136** are removed and a special lid **140** is installed and locked in place. Rats carry all kinds of diseases. A user or caretaker for the trap assembly **130** should avoid handling the captured rats as much as possible. The lid **140** on the trap assembly **130** is designed to allow the euthanasia process to take place while the rats are in the trapping chamber **138**. Once euthanized, the rats may be dumped into a container suitable for incineration. The trapping chamber **138** may be sanitized and placed back in service. It is envisioned this trap assembly **130** may be used in cities and towns everywhere. It can be installed just about anywhere, for example by dumpsters, building, gardens, etc.

[0084] The trap assembly **130** can be enlarged to provide access from all four sides (see FIGS. 23-24). In this case, a detachable bait cone **142** may be positioned in the center of the assembly **130**. The bait cone **142** is an upside down funnel with bait in the center thereof. The rat will jump to get to the bait only to slide down into the interior of the cone **142** and in to the trapping chamber **138**. The cone **142** may project upwardly from the bottom of the trapping chamber **138** and be removable where the trap assembly **130** is being processed, cleaned, replaced, etc. The trap assembly **130** may be used in combination with the trap **10** to create a self-baiting trap to bait for pythons or other target animals. The inventors are not aware of another trap designed to capture one animal (i.e., rats with the trap assembly **130**) to be used as bait for another animal (i.e., pythons in the trap **10**) all in the same trap complex.

[0085] The trap assembly **130** could be attached/mounted inside the auxiliary cage **70** of the trap **10** and is designed to capture dozens of rats and be durable enough to withstand the efforts of the pythons or other invasive species to get to the rats to eat them. The larger the heat signature coming from the trap will only extend the reach of the trap and attract more. Top portion **138** of the rat trap assembly **130** could be made of solid material so that the rats can't climb into the bait and entrance portion of the trap **130** and to support these components. The bottom portion **132** of the trap assembly **130** may be constructed of screen, perforated metal, or wire mesh to allow for the noise, scent, and heat to radiate outward. The entrance **136** may be a tube or pipe with a one-way angled door at the end, so as the rat pushes through the door it will close behind them and they can only move forward. The bait will be in a free spinning perforated/screen tube extending across the top of the trap **130**. The bait placement is designed to get the scent of the rat bait in the air, like peanuts. But the tube will not allow the rats access to eat the bait.

[0086] FIG. 25A shows a first option which allows the rats to jump to the bait and thus fall off because the structure supporting the rats spins. A second option is shown in FIG. 25B which includes a small plank for the rats to perch on while they decide what they want to do, but the door has closed behind them to capture them in the trap assembly **130**. FIG. 25C shows another option which includes a slide that extends across all entrances so that as the rats step out on the slide to reach the bait, the slide gets steeper until they can no longer hold traction. The trap assembly **130** may include multiple entrance tubes. Further, a water bowl may be provided in the bottom for humane treatment of the trapped rats. A lid/top is provided on the trap assembly **130** to provide access to the bait and captured rats.

[0087] Referring to FIGS. 26-28, another embodiment of an animal trap **200** according to this invention is shown for capturing nuisance animals, and more particularly, python snakes **202** (FIG.

28). According to this embodiment, a first inner trap **204** is located within a second outer trap **206**. The trap **200** is a live trap and does not harm the animals captured therein. Specifically, according to this embodiment, the inner trap **204** attracts bait animals **208** such as rats, mice and the like while the outer trap **206** captures the target animal such as python snakes **202**. The bait animal **208** cannot be harmed by the target animal **202** while the bait animal **208** is within the inner trap **204**. Both traps **204**, **206** can be either a single capture (one of each animal) or a multiple capture where the respective trap **204**, **206** can catch several of each animal **208**, **202**, or a single bait animal **208** can attract multiple target animals **202**. The inner first trap **204** can be reconfigured as necessary based on the type and size of bait animal **208** being targeted.

[0088] The inner and outer traps **204**, **206** may be of any size and configuration within the scope of this invention. The inner trap **204** defines an inner trap containment volume **204a** and the outer trap **206** defines an outer trap containment volume **206a**. As seen best in FIGS. 26-27, in one embodiment the inner trap **204** has a cylindrical sidewall **216**, a generally circular top wall **218** and a trap door **214** on the top wall **218** for one way entry of the target animals **208** for the inner trap **204** such as rats, mice, rodents and other animals. A ramp **212** or other structure may assist the target animals **208** in reaching the trap door **214** for entry into the trap **204**. The outer trap **206** may have a cuboid shape with one or more side walls **220** and a top wall **222**. A one-way entry door **224** may be provided in one or more sections to allow for entry, but not exit, of the animals **208**, **202** into the outer trap containment volume **206a**. The one-way entry door **224** may be of a design as previously disclosed in other embodiments of this invention or of another design. The outer trap **206** may include an entry portal section in the form of a chute **226** which is in communication with the containment volume **206a** of the trap **206**. While the chute **226** and associated entry door **224** are located on one side of the trap **206**, it will be understood by those of ordinary skill in the art that entry doors **224** and/or chutes **226** may be found on as many sides of the trap **206** as desired. In various embodiments of this invention, the inner trap **204** is entirely located within the containment volume **206a** of the outer trap **206**.

[0089] In one mode of operation, rat bait **210** may be placed in the bottom/center of the inner trap **204** which will attract the target animals **208**, in one case rats. The rats **208** will naturally look for a way into the outer trap **206** and ultimately the inner trap **204** to get to the bait **210**. The rats **208** can scale the sides of the wire mesh cylinder or climb the sides of the trap **204** or use a ramp **212** to access the top of the inner trap **204**. From the top of the trap **204**, the rats **208** will fall into the interior volume **204a** of the trap **204** via a trap door **214** and associated slide **214a**. A solid lid **216** is provided on the trap **204** so the rats **208** can't climb to the top of the trap **204** to try to escape. The live bait animals **208** in the inner trap **204** may be rats, rodents, etc. become bait for the ultimate target animal **202**. The more rats **208** that are trapped in the inner trap **204**, the better. The rats **208** in the inner trap **204** attract the pythons **202** into the containment volume **206a** of the outer trap **206** via the one-way door **224**. Once in the outer trap **206**, the pythons **202** cannot access or eat the rats **208** in the inner trap **204** nor can the pythons **202** escape the containment volume **206a** of the outer trap **206** because the entry door does **224** not open from the containment volume **206a**.

[0090] The trap **200** of this embodiment may include a thermal base **124** underneath the trap **200**. The thermal base **124** may be integral to the trap **200** or an underlayment on which the trap **200** is supported. The base **124** may be a polymer or styrene composition that is black and absorbs heat from the daytime sun and provides a heat source at night. A black or dark color absorbs the heat and a thicker the base **124** will have greater thermal retention capability. Many of the target reptiles/snakes **202** are nocturnal and are attracted to heat on cool nights. The rodents **208** for bait are also nocturnal. The additional heat source will only help attract both, bait and target animals **208**, **202**.

[0091] While this invention has been described in conjunction with a number of embodiments, those skilled in the art will recognize that certain modifications to the described embodiments still fall within the spirit and scope of the invention. Accordingly, the scope of this invention is not

meant to be limited by the disclosure herein, but may be modified while maintaining its novel process and construction, as would be apparent to one of ordinary skill in the art.

Claims

1. A trap for a target animal, the trap comprising: a first inner trap defining a first inner trap containment volume adapted to contain a bait animal therein; a second outer trap defining a second outer trap containment volume adapted to contain the target animal therein; a one-way entry door in the second outer trap allowing for entry of the bait animal and the target animal therethrough and into the second outer trap containment volume and inhibit exit from the second outer trap containment volume once therein; a door on the first inner trap allowing for entry of the bait animal into the first inner trap containment volume while inhibiting entry of the target animal into the first inner trap containment volume; wherein the bait animal being confined in the first inner trap containment volume attracts the target animal to enter the second outer trap containment volume via the one-way entry door and thereby trap the target animal in the trap.
2. The trap of claim 1 wherein the first inner trap is located within the second outer trap containment volume.
3. The trap of claim 1 wherein the one-way entry door is divided into a plurality of one-way entry door segments.
4. The trap of claim 1 wherein the one-way entry door further comprises: a plurality of downwardly oriented spaced tines.
5. The trap of claim 4 wherein each of the plurality of downwardly oriented tines is pivotally mounted about an axis spaced from the ground surface for independent pivotal movement relative to each other.
6. The trap of claim 4 wherein each of the plurality of downwardly oriented spaced tines is juxtaposed against a detent confronting the confinement volume.
7. The trap of claim 1 wherein the first inner trap has a generally cylindrical shape.
8. The trap of claim 1 wherein the door is located on a top portion of the first inner trap.
9. The trap of claim 8 further comprising: a ramp to provide the bait animal access to the door.
10. The trap of claim 1 wherein the first inner trap has a shape different from a shape of the second outer trap.
11. The trap of claim 1 wherein the second outer trap has a generally cuboid shape.
12. The trap of claim 1 further comprising: bait within the first inner trap containment volume to attract the bait animal thereto.
13. A trap for a target animal, the trap comprising: a first inner trap having a generally cylindrical shape and defining a first inner trap containment volume adapted to contain a bait animal therein; a second outer trap having a generally cuboid shape defining a second outer trap containment volume adapted to contain the target animal therein; a one-way entry door in the second outer trap allowing for entry of the bait animal and the target animal therethrough and into the second outer trap containment volume and inhibit exit from the second outer trap containment volume once therein; a door on the first inner trap located on a top portion of the first inner trap allowing for entry of the bait animal into the first inner trap containment volume while inhibiting entry of the target animal into the first inner trap containment volume; wherein the bait animal being confined in the first inner trap containment volume attracts the target animal to enter the second outer trap containment volume via the one-way entry door and thereby trap the target animal in the trap; wherein the first inner trap is located within the second outer trap containment volume.
14. The trap of claim 13 wherein the one-way entry door is divided into a plurality of one-way entry door segments.
15. The trap of claim 13 wherein the one-way entry door further comprises: a plurality of downwardly oriented spaced tines pivotally mounted about an axis spaced from the ground surface

for independent pivotal movement relative to each other and each of the plurality of downwardly oriented spaced tines is juxtaposed against a detent.

16. The trap of claim 13 further comprising: a ramp to provide the bait animal access to the door.

17. The trap of claim 13 further comprising: bait within the first inner trap containment volume to attract the bait animal thereto.

18. An animal trap comprising: at least one wall which at least in part defines a containment volume of the trap; a one-way entry door in the trap allowing for entry of the animal therethrough and into the containment volume and inhibit exit from the containment volume once therein; an escape door in the trap for allowing egress of a non-target animal in the containment volume from the containment volume, the escape door inhibiting egress of a target animal in the containment volume when in the containment volume.

19. The animal trap of claim 18 further comprising: a biasing member coupled to the escape door biasing the escape door toward a closed position.

20. The animal trap of claim 18 further comprising: a handle accessible to the non-target animal for movement of the escape door from a closed position toward an open position for egress from the containment volume.
